

ODAP Landscape 2026

The five-standard matrix for Open Data Agent
Platforms

Definition. An Open Data Agent Platform (ODAP) is a governed entry point where autonomous agents query heterogeneous data - open table formats, warehouses, object storage - through a semantic layer, with cost, permission, and audit controls enforced before execution.

Publisher note and disclaimer

Report type. This is a category-definition report for ODAP - Open Data Agent Platform. It is designed to frame a market, define evaluation criteria, and create a vocabulary for comparing agentic data platforms.

Independence. GoogleSQL.com is an independent project and is not affiliated with Google. The use of the word GoogleSQL refers to a product and research direction built around SQL and governed AI-to-data workflows; it does not imply endorsement by Google.

Scoring. Scores are editorial and directional, based on public product positioning as of the publication date. A score of 1 means weak or mostly absent; 3 means partial; 5 means strong and enterprise-relevant. The score is not a procurement recommendation, certification, benchmark, or legal opinion.

Scope. This report evaluates Databricks, Google, Starburst, Acceldata, and the generic category of Agent Cloud platforms. The Agent Cloud row does not refer to a single vendor; it refers to agent-first cloud infrastructure platforms whose primary role is runtime, orchestration, sandboxing, and persistent compute.

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1. Executive summary

The enterprise data platform market is entering a new phase. The previous competition was about who owned the fastest warehouse, the most complete lakehouse, the cleanest catalog, or the most governed BI layer. The 2026 competition is about who becomes the governed entry point through which autonomous agents access enterprise data.

That entry point is ODAP: Open Data Agent Platform. ODAP is not just Text-to-SQL. It is not merely a data catalog, a BI chatbot, an agent runtime, or an MCP server. ODAP is the governed valve between autonomous agents and enterprise data.

The valve narrative: as agents move from demos into production, enterprises need a choke point that can decide which agent may ask which question, against which data, through which semantic definition, at what cost, under which policy, with what audit trail. ODAP is that choke point.

In this landscape, Databricks and Google are closest to full ODAP readiness. Databricks has the strongest lakehouse-native route through Unity Catalog, Metric Views, Genie, Genie Agents, lineage, and AI governance. Google has the strongest cloud data-to-AI route through BigQuery, BigLake / Iceberg, Looker semantics, Knowledge Catalog, Gemini in BigQuery, and Gemini Enterprise Agent Platform. Starburst is the most compelling open-data access contender because its federated query and Trino-based architecture are naturally aligned with the Universal Data Gateway layer. Acceldata is best understood as a governance and observability control-plane specialist. Agent Cloud platforms are essential runtime infrastructure, but they are not complete ODAPs without data-native semantics, cost controls, and auditability.

The central conclusion is simple: the winner of ODAP will not be the tool that writes the most SQL. It will be the platform that governs which agent is allowed to run which query, against which data, under which semantic definition, before execution, with a complete evidence trail.

2. Why ODAP, why now

2.1 Autonomous agents are becoming data consumers

For decades, the primary data consumers in enterprises were analysts, dashboards, applications, and data pipelines. In 2026, agents are joining that list. They are not passive readers. They plan tasks, select tools, generate SQL, call APIs, inspect files, trigger workflows, and revise their own plans after errors. That makes them fundamentally different from a BI user clicking a filter or a data engineer running a scheduled job.

When an autonomous agent touches enterprise data, the organization needs to know whether the agent is authorized, whether the metric is correct, whether the query is too expensive, whether sensitive data is exposed, whether the result is explainable, and whether the decision trail can be reconstructed. These questions are not solved by giving the agent a database password or a generic connector.

2.2 Text-to-SQL was the demo; ODAP is the production architecture

Text-to-SQL solves an important interface problem: many users can express business questions more easily than they can write SQL. But production adoption fails when generated SQL is plausible but wrong, when it uses the wrong metric, when it scans too much data, when it leaks fields, when it cannot be audited, or when the answer changes because the model guessed a different join path.

ODAP elevates the problem from language generation to governed execution. It asks: how does a natural-language question become a semantic plan, a validated query, a policy decision, a cost estimate, a safe execution, an explainable result, and a durable audit record?

2.3 Open data is non-negotiable

Enterprise data is not located in one system. It lives in cloud warehouses, object storage, operational databases, SaaS applications, streaming systems, and open table formats such as Apache Iceberg and Delta Lake. Iceberg is explicitly designed as a high-performance table format that can be used safely by engines such as Spark, Trino, Flink, Presto, Hive, and Impala on the same tables [S15]. Delta Lake provides ACID transactions and scalable metadata handling on existing data lakes such as S3, ADLS, GCS, and HDFS [S16].

If a product only allows agents to query one warehouse, it may be a useful warehouse assistant, but it is not a complete Open Data Agent Platform. ODAP must be open enough to meet the data estate where it is.

3. Formal definition and category boundaries

Formal definition: An Open Data Agent Platform (ODAP) is a governed entry point where autonomous agents query heterogeneous data - open table formats, warehouses, object storage - through a semantic layer, with cost, permission, and audit controls enforced before execution.

3.1 What ODAP is

ODAP is an execution-path control plane. It sits between agents and data sources. It does not merely advise agents; it governs them. It does not merely discover tables; it decides whether a query should run. It does not merely explain SQL after the fact; it validates and controls the query before execution. It does not merely log that something happened; it records the full chain from intent to output.

3.2 What ODAP is not

ODAP is not a single-feature SQL assistant. It is not a BI chatbot limited to dashboard questions. It is not only a data catalog. It is not only data observability. It is not only an agent runtime. It is not only an MCP server. MCP is an important open standard for connecting AI applications to external systems, data sources, tools, and workflows [S17][S18], but it does not by itself define business metrics, estimate query cost, enforce warehouse permissions, or produce a regulatory audit package.

3.3 The category claim

ODAP is a new category because autonomous agents change the shape of the data-access problem. A human data analyst usually has context, judgment, and organizational accountability. An agent may have partial context, delegated authority, and high execution speed. Therefore the system must provide the missing controls: semantic grounding, pre-execution policy, cost limits, and auditability.

4. The ODAP five-standard matrix

The five standards are intentionally strict. They separate production ODAP platforms from adjacent tools. A platform can be excellent in one or two dimensions and still not be a complete ODAP.

What counts as ODAP?

Open Data Reach - The platform can reach heterogeneous enterprise data: open table formats, object storage, cloud warehouses, databases, and data products without forcing every workload into one proprietary store.

Semantic Layer - The platform translates business questions into governed metrics, dimensions, entities, time grains, relationships, and certified definitions before SQL or tool calls are executed.

Multi-Agent Pipeline - The platform uses a production workflow of planning, discovery, SQL generation, validation, correction, explanation, visualization, and memory rather than a single prompt-to-SQL step.

Pre-Execution Governance - Cost, permission, privacy, policy, budget, sandbox, and approval controls are enforced before the query or action actually runs.

End-to-End Auditability - The platform records the prompt, semantic resolution, generated SQL, policy decision, dry run, execution identifier, cost, output, user correction, and memory write-back.

Total Score Ranking



4.1 Detailed standard definitions

1. Open Data Reach

The platform can reach heterogeneous enterprise data: open table formats, object storage, cloud warehouses, databases, and data products without forcing every workload into one proprietary store.

2. Semantic Layer

The platform translates business questions into governed metrics, dimensions, entities, time grains, relationships, and certified definitions before SQL or tool calls are executed.

3. Multi-Agent Pipeline

The platform uses a production workflow of planning, discovery, SQL generation, validation, correction, explanation, visualization, and memory rather than a single prompt-to-SQL step.

4. Pre-Execution Governance

Cost, permission, privacy, policy, budget, sandbox, and approval controls are enforced before the query or action actually runs.

5. End-to-End Auditability

The platform records the prompt, semantic resolution, generated SQL, policy decision, dry run, execution identifier, cost, output, user correction, and memory write-back.

4.2 Why these five standards matter together

Open data reach without semantics produces uncontrolled access. Semantics without open data reach becomes a BI-only layer. Multi-agent workflow without pre-execution governance becomes an automation risk. Governance without auditability produces policies that cannot be proven. Auditability without semantic context records what happened but not whether it was correct.

ODAP requires all five standards because enterprise data agents must be both capable and accountable. The platform must help agents find data, understand data, plan queries, enforce rules, run safely, and prove what happened.

5. Methodology

This report uses a 1 to 5 scale for each of the five ODAP standards. The score reflects public evidence, product positioning, documentation, and product architecture fit. It is not a benchmark and does not test performance, latency, pricing, or feature completeness in a hands-on environment.

Score	Meaning	ODAP interpretation
5	Platform-native and enterprise-ready	The capability is a first-class product primitive and can be placed in the agent execution path.
4	Strong but bounded	The capability is mature, but has scope, ecosystem, or integration limits.
3	Partial	The capability exists but often depends on configuration, partners, or adjacent tools.
2	Adjacent or early	The platform has related primitives, but not enough to satisfy the ODAP standard.
1	Mostly absent	The platform usually requires an external product or custom engineering to satisfy the standard.
0	No public evidence	No credible public signal was found.

5.1 Evidence hierarchy

The report prioritizes official documentation, official product pages, official blogs, and major press releases. Third-party commentary is used only for general category context. When a platform announces broad capabilities but does not expose an execution-path product pattern, the report scores conservatively.

5.2 Scoring philosophy

A high ODAP score requires more than strong AI branding. The platform must have data-native controls. Conversely, a platform can receive a high score in open data reach or auditability without being a full ODAP if it lacks semantic or multi-agent execution depth. This is why Starburst scores very high in open data reach and auditability but lower in semantic layer and multi-agent pipeline. It is also why Agent Cloud platforms score high in multi-agent pipeline but low in data-native governance.

6. ODAP Landscape 2026 scoring matrix

The matrix below scores five representative platforms or categories against the ODAP five-standard model.

Vendor	Open Data Reach	Semantic Layer	Multi-Agent Pipeline	Pre-Execution Governance	End-to-End Auditability	Total / 25	ODAP Readiness
Databricks	5	4	4	5	5	23	Leader - Closest to full ODAP
Google	4	4	4	5	5	22	Leader - Strong in governance and semantics
Starburst	5	2	2	4	5	18	Open-data contender - Strong on open data
Acceldata	4	1	2	4	5	16	Control-plane specialist - Strong in observability and governance
Agent Cloud*	2	1	5	2	2	12	Agent-first, not ODAP-complete

Scores are editorial and directional. Agent Cloud refers to a generic platform category, not one vendor.

7. Market interpretation

7.1 Full-stack ODAP leaders

Databricks and Google are full-stack contenders because they control large parts of the data plane, governance plane, semantic plane, and agent plane. Their challenge is not whether they have enough primitives; it is whether they can package those primitives into a clear ODAP product path.

7.2 Open Data Gateway contenders

Starburst is the archetype. Its value is not to own every downstream workflow, but to make distributed data queryable without centralization and without bypassing governance. In a future ODAP stack, Starburst can be the universal query gateway that agents call through an MCP or governed SQL endpoint.

7.3 Control-plane specialists

Acceldata is the archetype. It may not be the primary question-answering surface, but it can supply quality, lineage, anomaly detection, observability, runtime control, and policy evidence. In many enterprises, ODAP will not replace observability; it will depend on it.

7.4 Agent runtime platforms

Agent Cloud platforms are critical, but they solve a different layer. They run agents; ODAP governs how those agents touch data. A mature enterprise architecture will likely combine an agent runtime with an ODAP data-control layer.

8. Vendor deep dives

The following sections explain each score in detail.

Databricks: Lakehouse-native ODAP leader

ODAP readiness: Leader - Closest to full ODAP. **Total score:** 23 / 25. [\[S1\]](#) [\[S2\]](#) [\[S3\]](#) [\[S4\]](#) [\[S16\]](#)

Databricks is one of the closest platforms to a complete ODAP because its lakehouse foundation, Unity Catalog governance, Metric Views, Genie / Genie Agents, lineage, audit logs, and AI governance controls are increasingly part of one platform narrative.

Standard	Score	Detailed rationale
Open Data Reach	5	The 5 reflects a platform-native lakehouse foundation: Delta Lake, open-format orientation, cloud storage integration, and lakehouse governance are core to the Databricks operating model. The score is high even though the best experience is still Databricks-native.
Semantic Layer	4	The 4 reflects Metric Views, semantic metadata such as synonyms and display names, and Genie Ontology. The gap to 5 is cross-platform neutrality and whether the semantic contract is fully independent of a Databricks-centered stack.
Multi-Agent Pipeline	4	The 4 reflects Genie Agents, Agent Bricks, and an emerging agent framework. It is not yet scored 5 because the ODAP-specific query pipeline - cost, permission, sandbox, SQL correction, audit, memory - is not packaged as a single named category.
Pre-Execution Governance	5	The 5 reflects Unity Catalog access control, attribute-based policies, row and column filters, lineage, and AI governance. These are strong ingredients for pre-execution enforcement.
End-to-End Auditability	5	The 5 reflects lineage, audit logs, system tables, and governance metadata that can map query actions to downstream assets and compliance evidence.

Strategic strengths

- Deep lakehouse and open-format foundation around Delta and Iceberg-compatible ecosystems.
- Unity Catalog gives strong access control, lineage, discovery, and AI governance primitives.
- Metric Views and Genie Ontology move the semantic layer closer to the data layer.
- Genie Agents and Agent Bricks give Databricks a credible route from conversational analytics to autonomous data agents.

ODAP gaps and watch items

- The experience is strongest inside the Databricks estate; truly neutral multi-platform ODAP use cases may still require external gateways.
- The product story is broad; buyers may need help separating general Data Intelligence Platform capabilities from ODAP-specific capabilities.

Buyer interpretation

Best fit for organizations already building a lakehouse-centered data and AI operating model. A buyer should evaluate how much of the intended ODAP traffic lives inside Databricks versus external warehouses and applications.

What would raise the score?

A move from leader to category-defining ODAP would require a clearer, cross-platform ODAP surface with an explicit agent-to-data policy contract, not merely a collection of strong data and AI capabilities.

Google: Cloud data-to-AI ODAP leader

ODAP readiness: Leader - Strong in governance and semantics. **Total score:** 22 / 25. [\[S5\]](#) [\[S6\]](#) [\[S7\]](#) [\[S8\]](#) [\[S17\]](#) [\[S18\]](#)

Google has a powerful ODAP route through BigQuery, BigLake / Iceberg, Looker semantics, Knowledge Catalog, Gemini in BigQuery, and the Gemini Enterprise Agent Platform. Its strength is the depth of governance, cost control, semantic grounding, and agent platform primitives in Google Cloud.

Standard	Score	Detailed rationale
Open Data Reach	4	The 4 reflects BigQuery, BigLake, Iceberg managed tables, and strong open-format momentum. It is not a 5 because the most polished operating model is still Google Cloud-centered rather than vendor-neutral.
Semantic Layer	4	The 4 reflects Looker semantics, Knowledge Catalog, BigQuery data canvas, and semantic governance direction. The gap is that the full semantic-to-agent path is distributed across multiple products.
Multi-Agent Pipeline	4	The 4 reflects Gemini Enterprise Agent Platform, ADK, MCP, A2A, agent registry, identity, gateway, and observability. It is not yet a 5 for ODAP because the data-query-specific agent pipeline is still assembled from several surfaces.
Pre-Execution Governance	5	The 5 reflects IAM, BigQuery dry runs and cost controls, audit logs, and semantic governance policies that allow controls to be applied before agent actions run.
End-to-End Auditability	5	The 5 reflects Cloud Audit Logs, BigQuery audit logs, catalog metadata, access logs, and governance workflows that can support a full agent-to-data evidence chain.

Strategic strengths

- BigQuery is a mature governed data platform with cost and audit primitives useful for agentic query control.
- BigLake and Iceberg managed tables strengthen open-format lakehouse support.
- Looker and semantic governance policies give Google a credible semantic grounding path for enterprise agents.
- Gemini Enterprise Agent Platform introduces agent registry, identity, gateway, observability, and orchestration concepts that match ODAP control-plane needs.

ODAP gaps and watch items

- The ODAP experience is distributed across BigQuery, Looker, Knowledge Catalog, and Gemini Enterprise; a buyer still has to assemble the full path.
- The platform is strongest inside Google Cloud, while neutral cross-cloud data access may be better served by federation-first products.

Buyer interpretation

Best fit for organizations already using BigQuery, Looker, Google Cloud governance, and Gemini-based enterprise agents. A buyer should evaluate the amount of integration work needed to make BigQuery, Looker, Knowledge Catalog, and Gemini feel like one governed ODAP surface.

What would raise the score?

A move from leader to category-defining ODAP would require a clearer, cross-platform ODAP surface with an explicit agent-to-data policy contract, not merely a collection of strong data and AI capabilities.

Starburst: Open Data Gateway contender

ODAP readiness: Open-data contender - Strong on open data. **Total score:** 18 / 25. [\[S9\]](#) [\[S10\]](#) [\[S11\]](#) [\[S15\]](#) [\[S17\]](#)

Starburst is the strongest open-data access contender in this landscape. Its Trino-based federated query model, Iceberg orientation, query-in-place pattern, and data product concept map naturally to the ODAP Universal Data Gateway. Its gap is less about data reach and more about agent-native semantics and multi-agent execution.

Standard	Score	Detailed rationale
Open Data Reach	5	The 5 reflects Starburst' s core strength: federated query, Trino, query-in-place, hybrid deployment, Iceberg optimization, and access to data across warehouses, lakes, databases, and applications.
Semantic Layer	2	The 2 reflects data products and semantic abstraction, but not a complete metric-centric semantic layer with governed KPIs, time grains, and agent-readable definitions.
Multi-Agent Pipeline	2	The 2 reflects MCP and AI-agent adjacency, but Starburst is not primarily a multi-agent SQL generation and correction platform.
Pre-Execution Governance	4	The 4 reflects query governance, access controls, policies, and workload management. The missing 5 is deeper agent-specific policy, prompt-aware governance, and SQL cost-before-execution as a unified workflow.
End-to-End Auditability	5	The 5 reflects query audit, query history, lineage-oriented data products, and the value of a single federated query trace across distributed data.

Strategic strengths

- Federated access to warehouses, lakes, databases, and applications makes Starburst highly relevant to open data agents.
- Trino and Iceberg orientation reduce the need to centralize all data before it can be queried.
- Data products can package curated assets with context, ownership, access policies, and quality expectations.
- MCP support gives agents a governed SQL endpoint rather than one-off custom connectors.

ODAP gaps and watch items

- The semantic layer is more data-product-centric than metric-centric; it is not yet a full business semantic layer.

- The multi-agent query workflow is lighter than Databricks or Google; Starburst is more gateway than full agent execution platform.

Buyer interpretation

Best fit for organizations whose data is distributed and cannot be centralized quickly. A buyer should treat Starburst as the open data gateway layer and pair it with a deeper semantic and agent orchestration layer when needed.

What would raise the score?

Starburst can raise its score by deepening a business semantic layer, adding a more explicit multi-agent SQL pipeline, and making prompt-to-query governance first-class.

Acceldata: Governance and observability control plane

ODAP readiness: Control-plane specialist - Strong in observability and governance. **Total score:** 16 / 25. [\[S12\]](#) [\[S13\]](#)

Acceldata is not a full ODAP query entry point, but it is an important control-plane candidate. Its value is data observability, quality, lineage, runtime control, governance, secure access, and hybrid visibility across data workloads and agents.

Standard	Score	Detailed rationale
Open Data Reach	4	The 4 reflects hybrid visibility and control across data estates, not a query gateway. It sees many places where data lives, but does not itself function as the primary universal SQL entry point.
Semantic Layer	1	The 1 reflects the absence of a full business semantic layer. Metadata, lineage, and quality are necessary, but they do not define revenue, active users, churn, or net retention for agentic queries.
Multi-Agent Pipeline	2	The 2 reflects agentic data management and automated governance workflows, not a full natural-language-to-SQL multi-agent query engine.
Pre-Execution Governance	4	The 4 reflects governance-by-design, runtime control, quality checks, and secure access. It does not receive 5 because enforcement is not always inside the query execution path.
End-to-End Auditability	5	The 5 reflects observability, lineage, data quality, operational traces, and governance evidence across complex estates.

Strategic strengths

- Strong fit for quality, lineage, runtime control, and operational governance across data estates.
- Hybrid-by-default positioning maps to enterprises that cannot move everything into one cloud data platform.
- Can become the observability and policy evidence layer behind ODAP deployments.

ODAP gaps and watch items

- It is not primarily a natural language query gateway.
- It lacks a full business semantic layer and does not own the main SQL generation path.
- It is strongest as an ODAP companion layer rather than the ODAP entry point itself.

Buyer interpretation

Best fit for data platform teams that need observability, quality, governance evidence, and runtime control across hybrid estates. A buyer should position it as ODAP control plane rather than the complete query entry point.

What would raise the score?

Acceldata can raise its score by moving from observability and control-plane governance into a direct, governed query execution path with semantic resolution.

Agent Cloud*: Agent runtime and orchestration substrate

ODAP readiness: Agent-first, not ODAP-complete. **Total score:** 12 / 25. [\[S14\]](#) [\[S17\]](#) [\[S18\]](#)

Agent Cloud platforms are essential to the agentic stack, but they are not complete ODAPs. Their strength is runtime: sandboxes, persistent compute, background workers, storage, GPU access, tool orchestration, and long-running execution. Their weakness is data-native governance.

Standard	Score	Detailed rationale
Open Data Reach	2	The 2 reflects connector and tool access to data, but not open-format-native data reach. Agent clouds typically call APIs or tools rather than understanding Iceberg manifests, Delta logs, partitions, statistics, and query optimizers.
Semantic Layer	1	The 1 reflects the general absence of business semantic layers. Agents can call a semantic layer, but the cloud runtime usually does not own one.
Multi-Agent Pipeline	5	The 5 reflects the core strength of agent clouds: orchestration, sandboxes, long-running processes, background workers, state, tool routing, and retries.
Pre-Execution Governance	2	The 2 reflects sandbox and tool permission primitives, but not data-native pre-execution governance such as row-level access, cost estimation, sensitive-output masking, and SQL dry runs.
End-to-End Auditability	2	The 2 reflects agent traces, but not data-grade auditability. Tool traces do not prove metric definitions, query lineage, or warehouse cost attribution.

Strategic strengths

- Excellent for multi-agent orchestration, code execution, workflow state, and tool calling.
- Useful for running agents that call ODAP endpoints, BI tools, MCP servers, or warehouse APIs.
- Can provide the compute substrate for autonomous workflows.

ODAP gaps and watch items

- Usually lacks open-table-format understanding, SQL cost estimation, row/column-level access semantics, and data lineage.
- Tool traces are not the same as data audit trails.

- Without an external semantic layer and governed query gateway, agent clouds can become a shadow-data-access risk.

Buyer interpretation

Best fit for engineering teams building agents that need sandboxes, persistent compute, orchestration, and tool execution. A buyer should add a governed data gateway and semantic layer before allowing agents to reach enterprise data.

What would raise the score?

Agent Cloud platforms can raise their score only by adding data-native semantics, SQL-specific cost and permission controls, and warehouse-grade auditability rather than generic tool traces.

9. Horizontal comparison by ODAP standard

9.1 Open Data Reach

Starburst and Databricks score highest because they are structurally aligned with open data. Starburst is strongest in federated query and query-in-place across many systems. Databricks is strongest in lakehouse-native open-format workloads. Google is close through BigQuery, BigLake, and Iceberg managed tables, but the operating model remains Google Cloud-centered. Acceldata has hybrid reach as a control plane, not as a query gateway. Agent Cloud platforms connect to data through tools but do not understand open table formats natively.

9.2 Semantic Layer

Databricks and Google lead because they are building semantic assets close to governed data. Databricks has Metric Views and Genie Ontology; Google has Looker semantics, Knowledge Catalog, and semantic governance direction. Starburst has data products that can serve as semantic abstractions, but they are not yet a full metrics layer. Acceldata and Agent Cloud platforms usually rely on external semantic systems.

9.3 Multi-Agent Pipeline

Agent Cloud platforms lead in orchestration mechanics. Databricks and Google are close because their agent platforms are increasingly integrated with data platforms. Starburst and Acceldata have agent-adjacent capabilities, but they are not primarily multi-agent SQL engines.

9.4 Pre-Execution Governance

This is the decisive ODAP standard. Databricks and Google score 5 because their governance and cost-control primitives can be placed before execution. Starburst and Acceldata score 4 because they have strong governance or runtime control, but the agent-specific query gate is less complete. Agent Cloud platforms score lower because sandbox permissions are not the same as data-native access, cost, and privacy enforcement.

9.5 End-to-End Auditability

Traditional data platforms have strong audit logs, but ODAP requires a new audit object: the agent decision trail. The most mature platforms will connect prompt, identity, semantic definition, SQL, policy decision, cost estimate, execution ID, lineage, output, and memory update. Databricks,

Google, Starburst, and Acceldata all have strong audit ingredients; Agent Cloud platforms typically have traces but not full data audit trails.

10. ODAP reference architecture

A complete ODAP can be described as five product layers. These layers map directly to the five standards and to the product design originally proposed for GoogleSQL.com.

Layer	Purpose	Core capabilities	ODAP standard
Universal Data Gateway	Reach heterogeneous data	Warehouse connectors, object storage, Iceberg, Delta, Hudi, schema discovery, statistics, metadata sync, federated query	Open Data Reach
Smart Semantic Layer	Turn business language into governed semantics	Metrics, dimensions, entities, time grains, synonyms, certified definitions, semantic lineage	Semantic Layer
Multi-Agent SQL Engine	Convert intent into safe query execution	Planning, discovery, SQL generation, validation, cost estimation, policy check, correction, explanation	Multi-Agent Pipeline
Persistent Memory	Make the platform learn safely	User corrections, verified query patterns, business terms, rejected joins, quality caveats, permission-aware memory	Pipeline + Auditability
Governance & Observability Center	Control and prove agent-to-data operations	Audit logs, budgets, approvals, sandbox validation, quality scores, PII logs, lineage, incident replay	Governance + Auditability

10.1 The execution path

1. User or agent submits a natural-language question.
2. ODAP resolves identity, purpose, and session context.
3. The planner identifies candidate data sources and semantic objects.
4. The semantic layer maps business terms to certified definitions.
5. The SQL engine generates candidate plans.
6. The validator checks syntax, joins, filters, and metric consistency.
7. The policy engine checks permissions, sensitive data, and allowed use.
8. The cost engine estimates scan size, compute cost, and budget impact.

9. The sandbox runs dry-run or sample validation.
10. Approved execution occurs.
11. The result, explanation, visualization, confidence score, and audit record are returned.
12. User corrections and safe memory updates are written back.

10.2 Why persistent memory must be governed

Memory is one of the most powerful but dangerous ODAP layers. It can remember preferred metrics, approved join paths, and user corrections. But it can also leak context across departments or retain sensitive data. ODAP memory must be scoped by identity, role, dataset, policy, and semantic version. The memory layer must itself be auditable.

11. Buyer scenarios and adoption

11.1 Business users

Business users want natural-language answers with charts, explanations, and trustworthy metrics. ODAP gives them self-service without letting them bypass semantic definitions or permissions.

11.2 Data analysts

Analysts want faster SQL generation, query explanation, optimization, and reusable metric definitions. ODAP turns individual query work into governed organizational knowledge.

11.3 Data engineers

Data engineers want connected sources, metadata, quality signals, lineage, and performance visibility. ODAP gives them a way to see how agents are actually using data.

11.4 Administrators and security teams

Administrators want permissions, budgets, PII protection, audit logs, approvals, and incident replay. ODAP gives them a single place to govern agent-to-data traffic.

11.5 Executives

Executives want AI adoption without data chaos. ODAP offers a practical answer: make agents useful while forcing them through controlled semantic and policy layers.

12. Risk model and governance controls

Risk	Failure mode	ODAP control
Semantic drift	The same business term is interpreted differently across teams.	Certified metrics, semantic versioning, owner approval, ambiguity detection.
Permission leakage	Agent combines data in ways that expose unauthorized fields or rows.	Agent identity, row/column security, purpose-based policies, output masking.
Cost explosion	Agent generates a query that scans excessive data.	Dry run, maximum bytes, budget guardrails, cost preview, approval thresholds.
Hallucinated SQL	Agent uses nonexistent tables, wrong joins, or incorrect filters.	Schema retrieval, semantic constraints, SQL validation, sandbox execution, correction loops.
Audit gaps	The organization cannot reconstruct why a query ran.	Prompt-to-SQL trace, semantic model version, policy decision log, execution ID, memory log.
Agent overreach	Agent moves from answering to taking action without approval.	Read-only default, progressive authorization, human-in-the-loop approvals, action scopes.

The main ODAP security principle is that agents should not receive raw, permanent, over-broad data credentials. Instead, they should receive governed capabilities: ask a question, request a semantic object, propose a query, request execution, and receive approved results. Every step should be mediated.

13. GoogleSQL.com beachhead strategy

GoogleSQL.com should not begin by claiming to be a complete universal platform. The beachhead is BigQuery / GoogleSQL users, because the workflow is concrete: generate SQL, explain SQL, optimize SQL, estimate cost, dry run, inspect schema, and create audit-ready query records.

13.1 Phase 1: AI SQL Workbench

- Schema-aware SQL generation
- SQL explanation and optimization
- BigQuery dry-run cost preview
- Query safety warnings
- Saved query history

13.2 Phase 2: Semantic SQL Copilot

- Metric glossary
- Dimension registry
- Certified query templates
- Business synonyms
- User correction memory

13.3 Phase 3: Governed Agent Query Gateway

- Agent identity
- Permission checks
- Policy rules
- Budget thresholds
- Audit package exports

13.4 Phase 4: Full ODAP

- Multi-warehouse access
- Object storage and open table formats
- MCP / A2A agent endpoints
- Cross-platform semantic layer
- Governance and observability center

The strategic move is to define the category before owning the whole stack. By publishing the Manifesto, Landscape, Readiness Index, and buyer checklist, GoogleSQL.com can become the scoreboard even before it becomes the full platform.

Appendix A: ODAP readiness checklist

Question	Why it matters
Can agents query open table formats, warehouses, and object storage through one governed entry point?	This tests Open Data Reach.
Are metrics, dimensions, time grains, and entities defined before SQL generation?	This tests Semantic Layer.
Does the system perform planning, validation, correction, and explanation rather than one-shot SQL generation?	This tests Multi-Agent Pipeline.
Are permission, cost, privacy, and sandbox checks enforced before execution?	This tests Pre-Execution Governance.
Can the organization reconstruct prompt, SQL, policy decision, execution ID, cost, output, and user correction?	This tests End-to-End Auditability.
Does the platform have agent identity and scoped data capabilities rather than shared credentials?	This tests production governance.
Can memory be scoped, audited, corrected, and deleted?	This tests safe personalization.
Can administrators set budgets, kill switches, approval thresholds, and sensitive-data policies?	This tests control-plane maturity.

Appendix B: Glossary

Term	Definition
ODAP	Open Data Agent Platform: a governed entry point for autonomous agents querying heterogeneous enterprise data through a semantic layer.
The valve	The policy and execution choke point between agents and enterprise data.
Semantic layer	A governed business abstraction over data: metrics, dimensions, entities, relationships, and definitions.
Open table format	A table format such as Iceberg, Delta Lake, or Hudi that enables data lakehouse interoperability.
Pre-execution governance	Controls that run before a query or action executes, such as permission, cost, PII, sandbox, and approval checks.
Agent identity	A first-class identity assigned to an autonomous agent, with its own scopes, policies, and audit trail.
Audit package	A reconstructable evidence bundle containing prompt, semantic path, SQL, policy decision, cost, execution ID, result, and corrections.

Sources and citation notes

Accessed July 4, 2026. Sources are used for factual product-positioning context. Scores remain editorial and directional.

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